Nombre: Lucas D. Jumbo G.

1. **Creacion tabla “turnos”**

	idTurn	Codigo	Descripcion	idVel
*	NULL	NULL	NULL	NULL

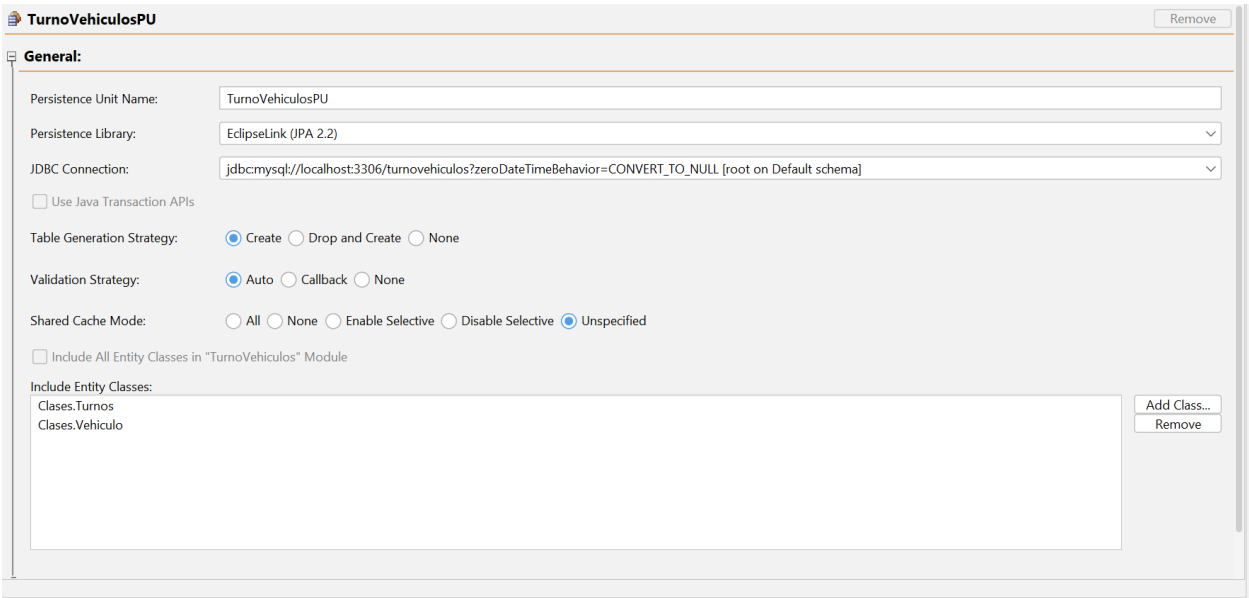
Se crea la table turnos con sus respecticas columnas, “Codigo” lo marque como un VARCHAR(45)

2. **Creacion tabla “vehiculo”**

	idVel	Year	Marca
*	NULL	NULL	NULL

Se crea la table vehículo, “idVel” es la PK

3. **Conexion persistence entre NetBeans y la BD**



Se realiza la conexion con “persistence.xml”

4. Crear el “Entity Class” de “turnos” desde la BD

```java
/**
 *
 * @author LucasJG
 */
@Entity
@Table(name = "turnos")
@NamedQueries({
    @NamedQuery(name = "Turnos.findAll", query = "SELECT t FROM Turnos t"),
    @NamedQuery(name = "Turnos.findByIdTurn", query = "SELECT t FROM Turnos t WHERE t.idTurn = :idTurn"),
    @NamedQuery(name = "Turnos.findByCodigo", query = "SELECT t FROM Turnos t WHERE t.codigo = :codigo"),
    @NamedQuery(name = "Turnos.findByDescripcion", query = "SELECT t FROM Turnos t WHERE t.descripcion = :descripcion"),
    @NamedQuery(name = "Turnos.findByIdVel", query = "SELECT t FROM Turnos t WHERE t.idVel = :idVel")})
public class Turnos implements Serializable {

    private static final long serialVersionUID = 1L;
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @Basic(optional = false)
    @Column(name = "idTurn")
    private Integer idTurn;
    @Basic(optional = false)
    @Column(name = "Codigo")
    private String codigo;
    @Basic(optional = false)
    @Column(name = "Descripcion")
    private String descripcion;
    @Basic(optional = false)
    @Column(name = "idVel")
    private int idVel;

    public Turnos() {
    }

    public Turnos(Integer idTurn) {
        this.idTurn = idTurn;
    }
```

5. Crear el “Entity Class” de “vehiculos” desde la BD

```java
/**
 *
 * @author LucasJG
 */
@Entity
@Table(name = "vehiculo")
@NamedQueries({
    @NamedQuery(name = "Vehiculo.findAll", query = "SELECT v FROM Vehiculo v"),
    @NamedQuery(name = "Vehiculo.findByIdVel", query = "SELECT v FROM Vehiculo v WHERE v.idVel = :idVel"),
    @NamedQuery(name = "Vehiculo.findByYear", query = "SELECT v FROM Vehiculo v WHERE v.year = :year"),
    @NamedQuery(name = "Vehiculo.findByMarca", query = "SELECT v FROM Vehiculo v WHERE v.marca = :marca")})
public class Vehiculo implements Serializable {

    private static final long serialVersionUID = 1L;
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @Basic(optional = false)
    @Column(name = "idVel")
    private Integer idVel;
    @Basic(optional = false)
    @Column(name = "Year")
    private int year;
    @Basic(optional = false)
    @Column(name = "Marca")
    private String marca;

    public Vehiculo() {
    }

    public Vehiculo(Integer idVel) {
        this.idVel = idVel;
    }

    public Vehiculo(Integer idVel, int year, String marca) {
        this.idVel = idVel;
        this.year = year;
```

6. Generar el controlador de “Vehiculo”

```java
package Logica;

import Clases.Vehiculo;
import Logica.exceptions.NonexistentEntityException;
import java.util.List;
import javax.persistence.EntityManager;
import javax.persistence.EntityManagerFactory;
import javax.persistence.EntityNotFoundException;
import javax.persistence.Persistence;
import javax.persistence.Query;
import javax.persistence.criteria.CriteriaQuery;
import javax.persistence.criteria.Root;

/**
 *
 * @author LucasJG
 */
public class VehiculoJpaController {
    public VehiculoJpaController() {
        this.emf = Persistence.createEntityManagerFactory("TurnoVehiculosPU");
    }
    private EntityManagerFactory emf = null;

    public EntityManager getEntityManager() {
        return emf.createEntityManager();
    }

    public void create(Vehiculo vehiculo) {
        EntityManager em = null;
        try {
            em = getEntityManager();
            em.getTransaction().begin();
            em.persist(vehiculo);
            em.getTransaction().commit();
        } finally {
            if (em != null) {
```

7. Generar el controlador de “Turnos”

```java
package Logica;

import Clases.Turnos;
import Logica.exceptions.NonexistentEntityException;
import java.io.Serializable;
import java.util.List;
import javax.persistence.EntityManager;
import javax.persistence.EntityManagerFactory;
import javax.persistence.EntityNotFoundException;
import javax.persistence.Persistence;
import javax.persistence.Query;
import javax.persistence.criteria.CriteriaQuery;
import javax.persistence.criteria.Root;

/**
 *
 * @author LucasJG
 */
public class TurnosJpaController implements Serializable {
    public TurnosJpaController() {
        this.emf = Persistence.createEntityManagerFactory("TurnoVehiculosPU");
    }
    private EntityManagerFactory emf = null;

    public EntityManager getEntityManager() {
        return emf.createEntityManager();
    }

    public void create(Turnos turnos) {
        EntityManager em = null;
        try {
            em = getEntityManager();
            em.getTransaction().begin();
            em.persist(turnos);
            em.getTransaction().commit();
        } finally {
            if (em != null) {
                em.close();
```

8. Codigo principal para insertar

```java
package turnovehiculos;

import Clases.Turnos;
import Clases.Vehiculo;
import Logica.TurnosJpaController;
import Logica.VehiculoJpaController;

/**
 *
 * @author LucasJG
 */
public class TurnoVehiculos {

    /**
     * @param args the command line arguments
     */
    public static void main(String[] args) {
        Vehiculo objVehiculo = new Vehiculo(1997, "Mazda");
        VehiculoJpaController objCtrV = new VehiculoJpaController();
        objCtrV.create(objVehiculo);

        Turnos objTurno = new Turnos("3456C", "Reparar Llanta", objVehiculo.getIdVel());
        TurnosJpaController objCtrT = new TurnosJpaController();
        objCtrT.create(objTurno);
    }
}
```

9. Run

```
run:
[EL Info]: 2026-04-21 11:48:37.31--ServerS
BUILD SUCCESSFUL (total time: 1 second)
```

El run del programa fue exitoso

10. Prueba de subida en “Vehiculo”

```
1 •  SELECT * FROM turnovehiculos.vehiculo
```

Result Grid | Filter Rows: | Edit:

	idVel	Year	Marca
▶	1	1997	MazdaPrueba
	2	1997	Mazda
*	NULL	NULL	NULL

11. Prueba de subida en “Turnos”

```
1 •  SELECT * FROM turnovehiculos.turnos
```

Result Grid | Filter Rows: | Edit:

	idTurn	Codigo	Descripcion	idVel
▶	1	3456C	Reparar Llanta	2
*	NULL	NULL	NULL	NULL

Como se ve en la imagen, “idVel” pertenece al vehículo “Mazda” de la tabla Vehiculos